AnnFlux: Object-conditioned neural stochastic differential equations for single-cell perturbation dynamics

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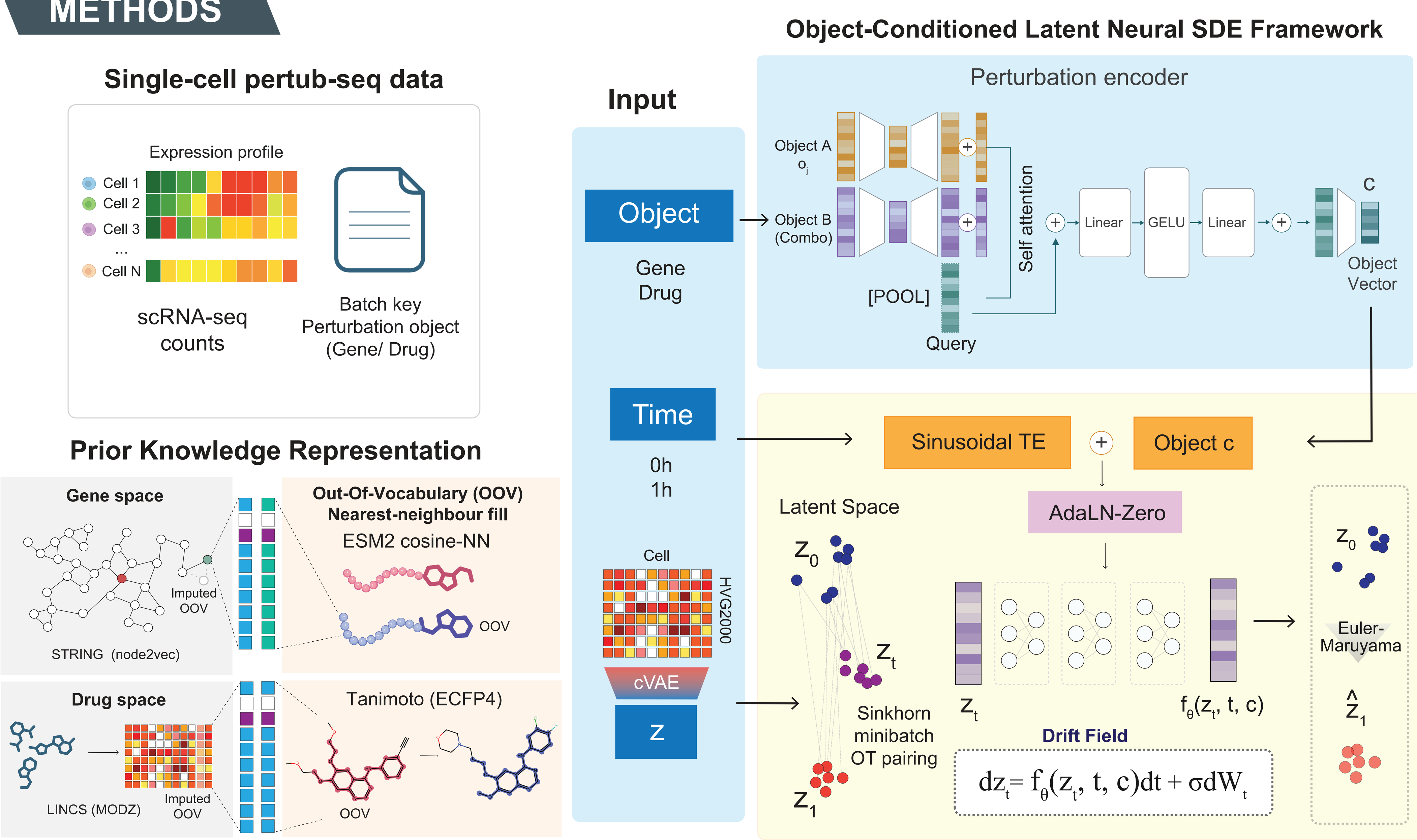
METHODS

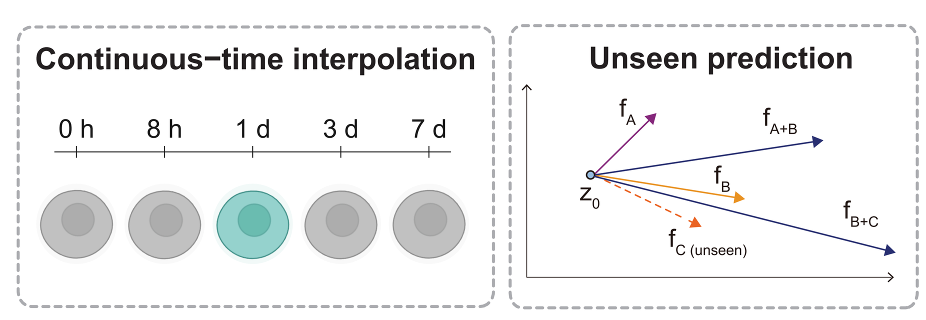


Pairing, training and inference

- Because true source–target cell correspondences are unobservable, Sinkhorn OT constructs surrogate pairs within groups sharing the same objects, cell line and time interval.
- Using these pairs, AnnFlux simulates source-to-target latent trajectories with 20-step Euler–Maruyama integration and learns the drift by matching simulated and observed endpoint populations with four loss terms; inference generates 10 trajectories per source cell using 30 steps.

Prior-knowledge object encoder

- Rather than learning embeddings solely from response data, the encoder uses STRING v12 Node2Vec embeddings for genes and LINCS-L1000 signatures for drugs, imputing missing objects from nearest neighbors in ESM2 protein-sequence or Morgan-fingerprint space, respectively.



Algorithm 1: AnnFlux training step (object-conditioned latent SDE)

Input : source (control) cells x_0, target cells x_1, perturbation objects $o = (o_1, \ldots, o_M)$; pair times τ_s, τ_e; train steps K=20, noise σ=0.5, weights (w_{mse}, w_{energy}=1, w_{scal}, w_{vel}=0.1)

Output: trained drift network u_θ

$e_j \leftarrow$ PKN-view(o_j)

$c \leftarrow$ Pool$(\{e_j\}_{j=1}^{M})$ `// [POOL] self-attention over object tokens → condition`

$x_1 \leftarrow$ Sinkhorn-OT re-pairing of (x_0, x_1) within each (o, τ) group (ε=0.05, 50 iters)

$\Delta t \leftarrow (\tau_e - \tau_s)/K; \quad x \leftarrow x_0$

for $k = 0$ **to** $K - 1$ **do**

  $t_k \leftarrow \tau_s + (k + \frac{1}{2})\,\Delta t; \quad \varepsilon \sim \mathcal{N}(0, I)$

  $x \leftarrow x + u_\theta(x, t_k, c)\,\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon$ `// Euler--Maruyama step`

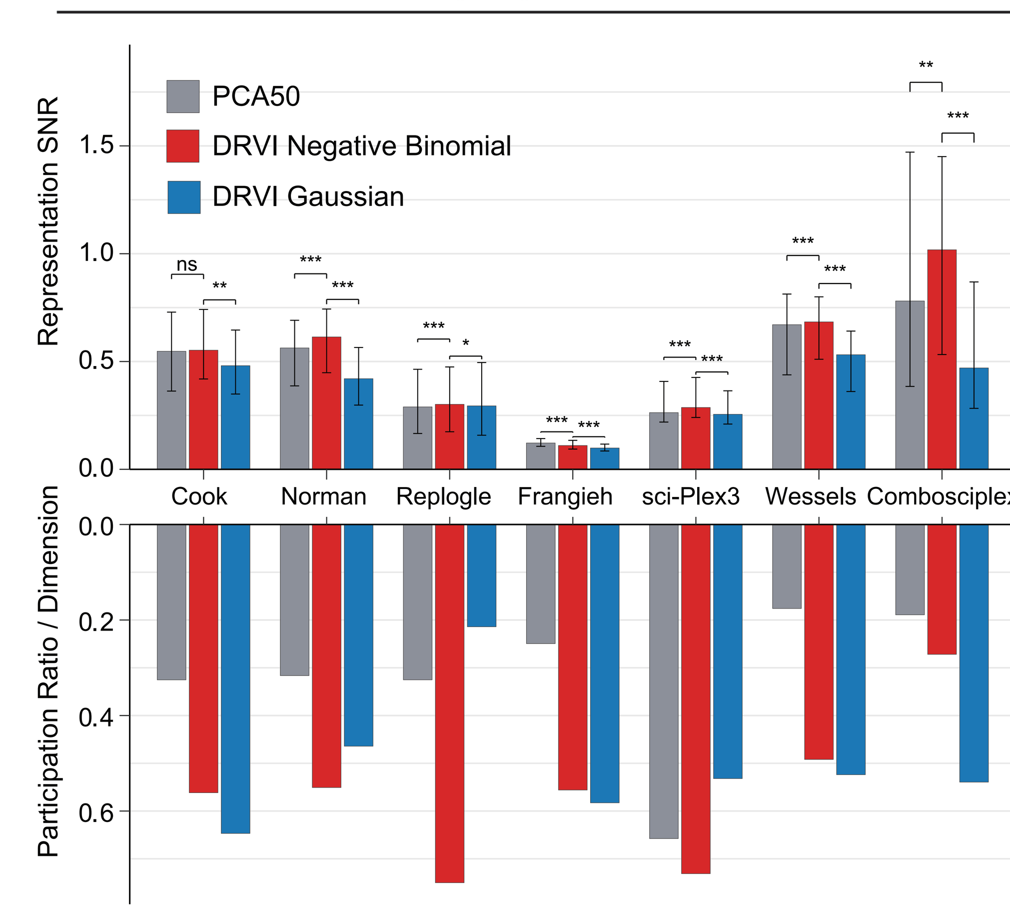
end

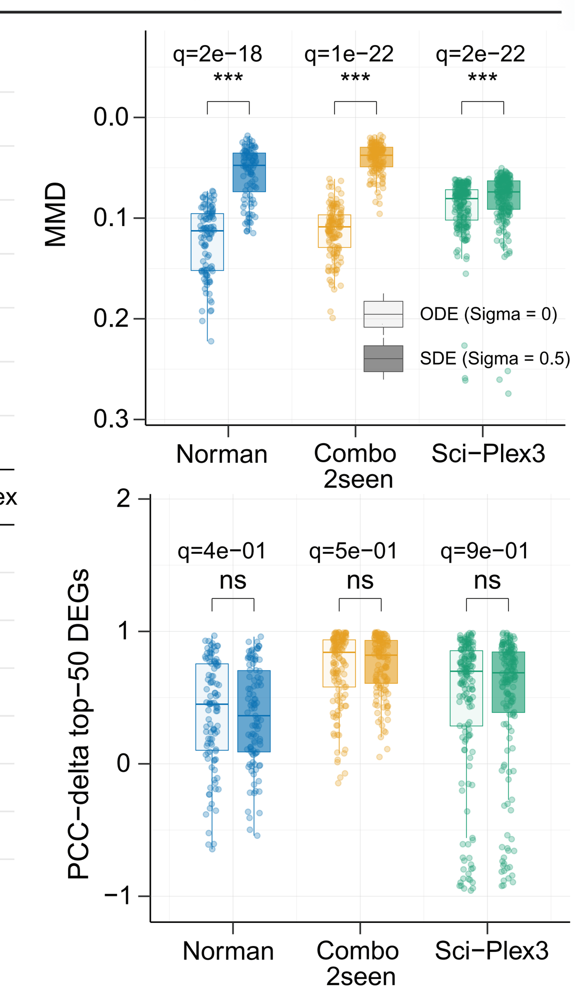
$\hat{x}_1 \leftarrow x$ `// simulated endpoint population`

$L \leftarrow w_{\text{mse}} L_{\text{MSE}} + w_{\text{energy}} L_{\text{energy}} + w_{\text{scal}} L_{\text{scalar}} + w_{\text{vel}} L_{\text{vel}}$ `// scalar-4-term`

$\theta \leftarrow$ AdamW$(\theta, \nabla_\theta L)$, lr 10^{-4}, wd 10^{-4}, grad-clip 1.0 `// cosine decay after ep. 10; early-stop on val loss, patience 30`

Inference: draw 10 samples per source cell with K=30 Euler–Maruyama steps.

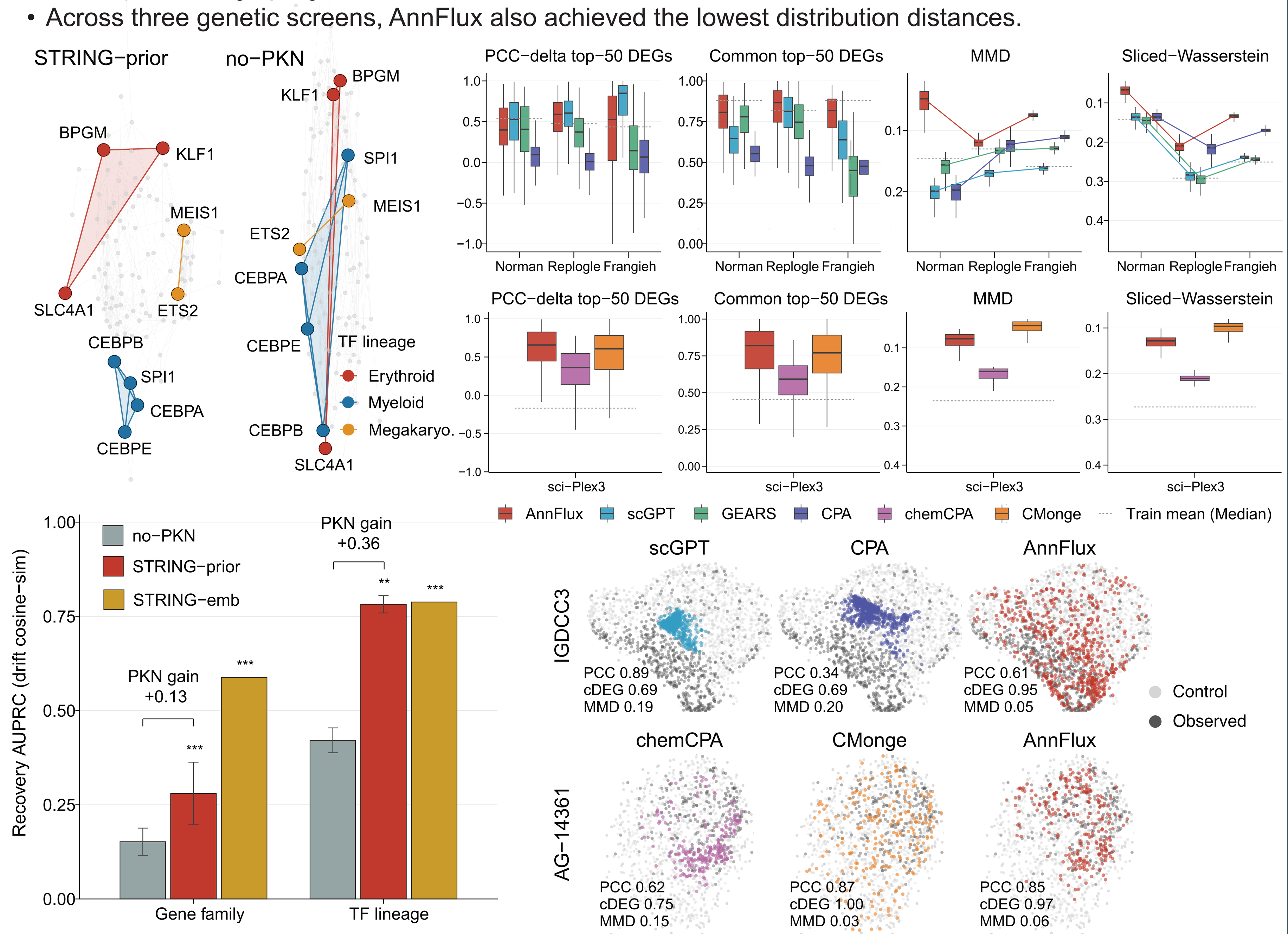




RESULTS

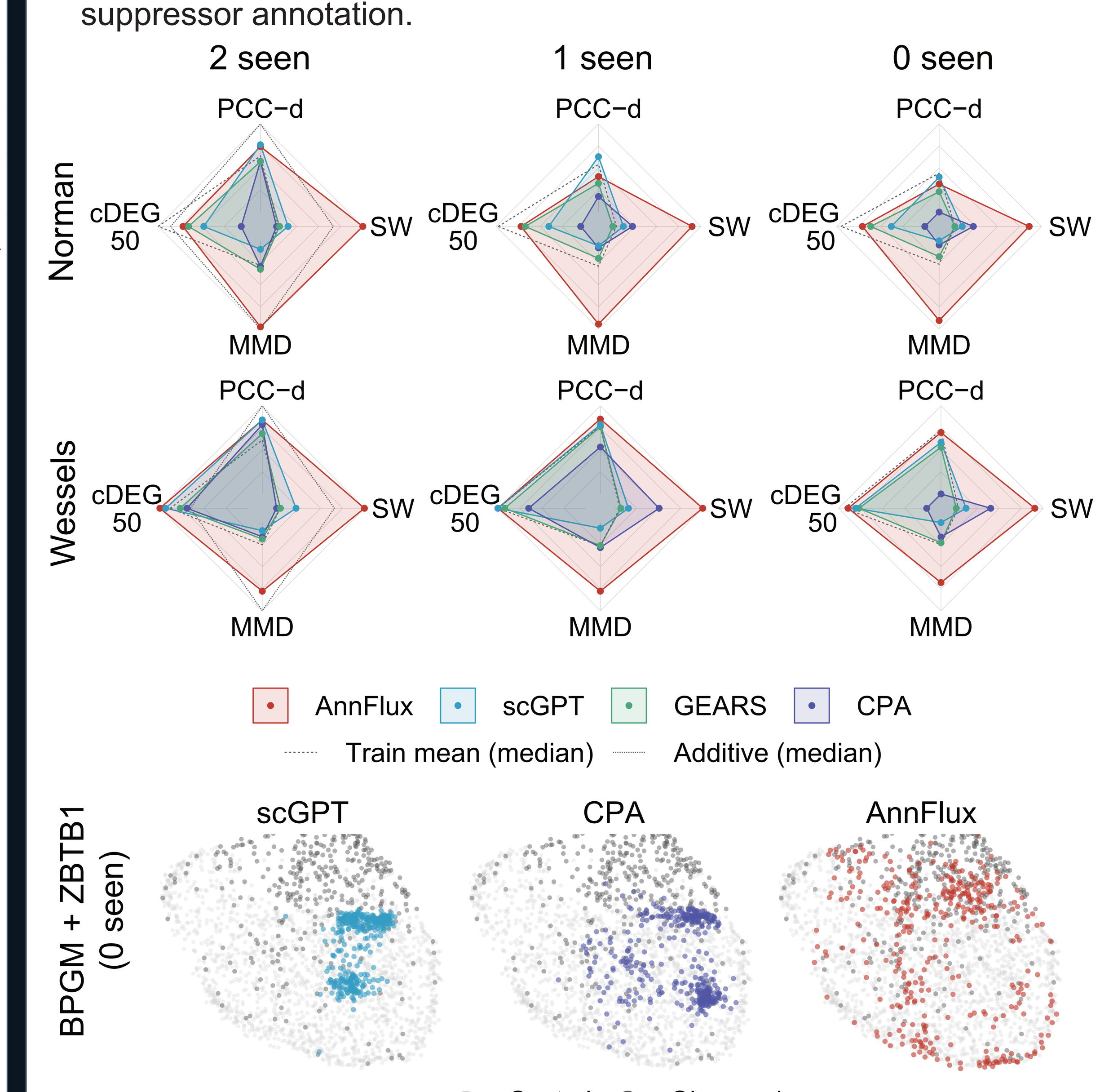
1. Predicting perturbations across genetic and chemical screens

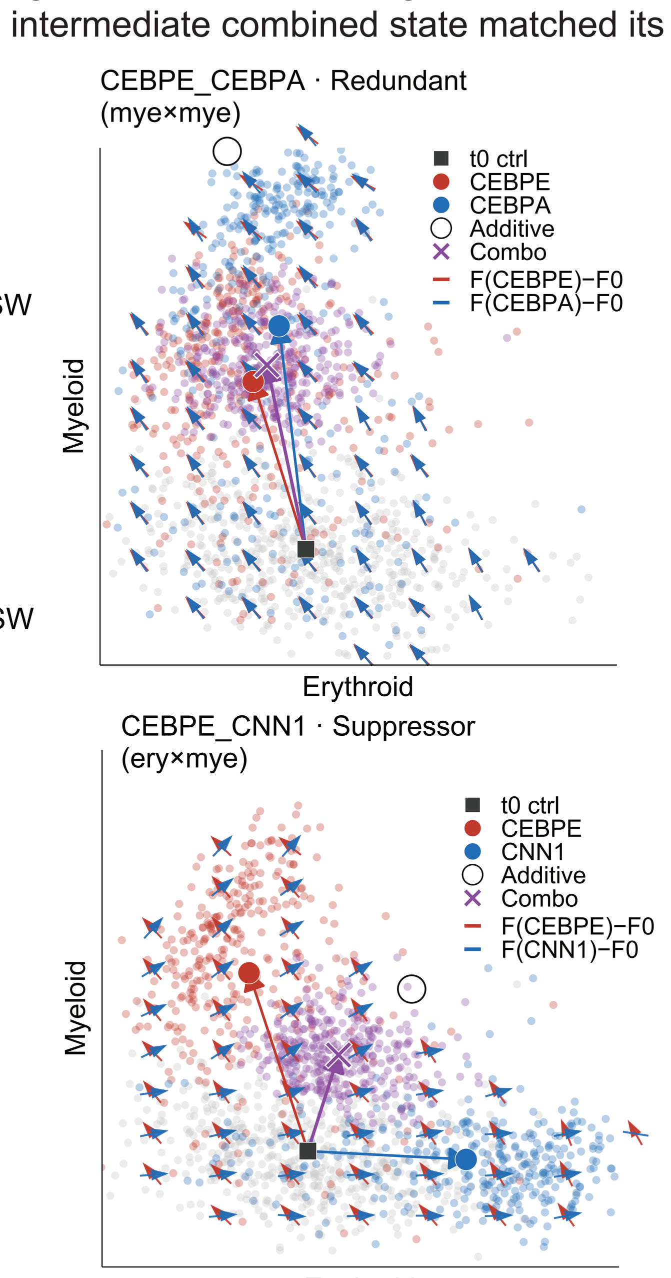
- A 4-NN network of endpoint-shift cosine similarities recovered lineage programs and gene modules, with within-module pair-recovery AUPRC increasing from 0.15 to 0.28 for HGNC gene families and from 0.42 to 0.78 for TF-lineage programs.
- Across three genetic screens, AnnFlux also achieved the lowest distribution distances.



3. AnnFlux predicts combinatorial perturbations and their interaction structure

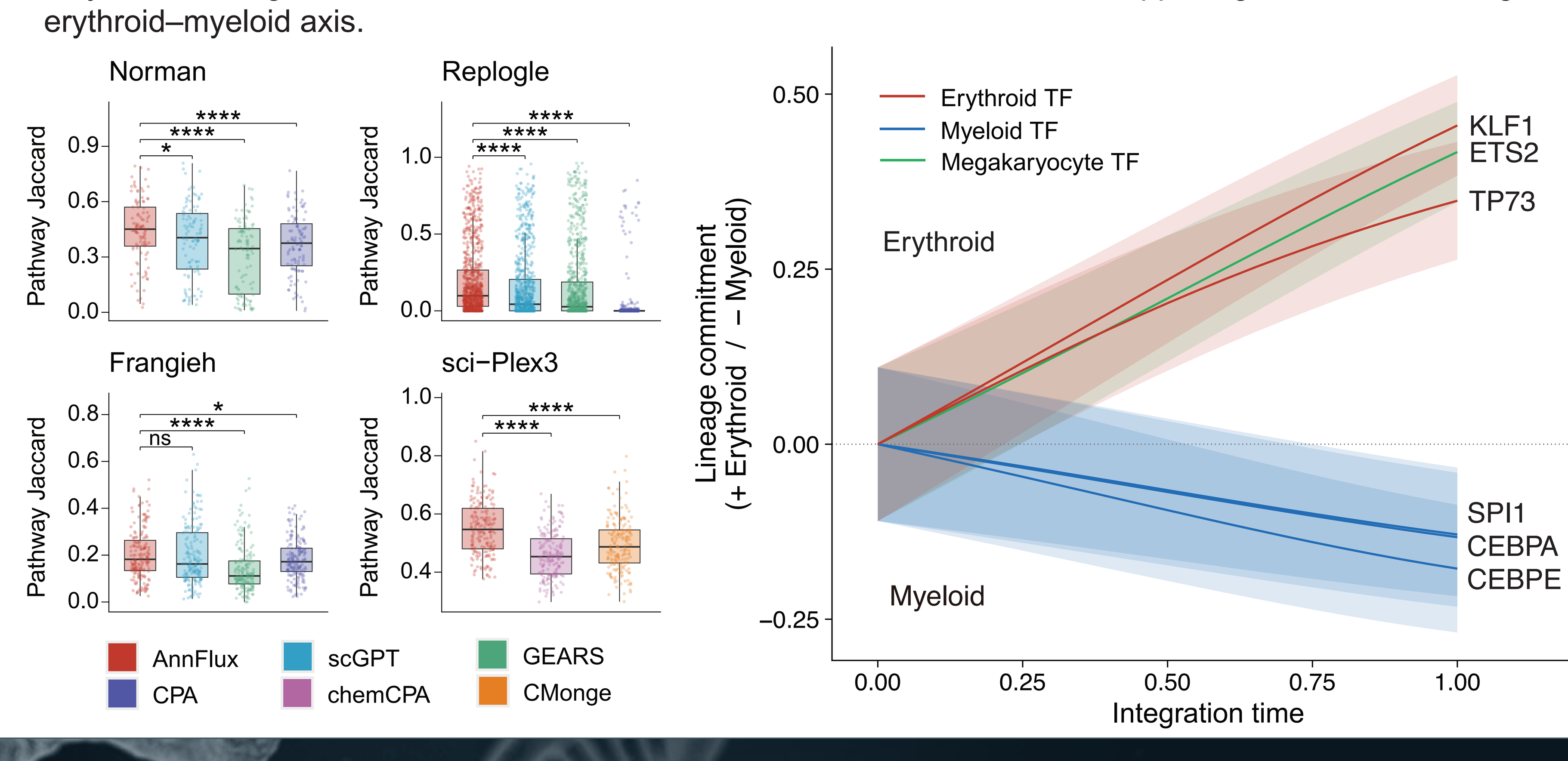
- Across all three regimes, AnnFlux gave the lowest distribution distances of the learned models. For representative held-out Norman combinations, AnnFlux recovered both the mean shift and cell-to-cell variation observed in the target population.
- Projecting Norman combinatorial drifts onto erythroid and myeloid program axes identified aligned, redundant CEBPE–CEBPA effects and opposing CEBPE–CNN1 effects whose intermediate combined state matched its suppressor annotation.





2. Drift decoding recovers cell-fate and gene-program organization

- AnnFlux achieved the highest overlap between predicted and observed enriched pathways in all four screens.
- Trajectories integrated from control cells for six held-out Norman TFs showed opposing commitment along the erythroid–myeloid axis.



HIGHRIGHT

The gap

- Existing perturbation models leave a key gap: CPA, GEARS and scGPT predict static responses, whereas scDiffEq and ARTEMIS learn continuous dynamics without perturbation-specific drift fields.

The solution

- AnnFlux closes this gap with an object-conditioned neural SDE that predicts both continuous cell-state trajectories and endpoint distributions. Its self-attention-based encoder enables the same framework to model combinatorial perturbations.

The evidence

- AnnFlux achieved its most consistent gains on distributional metrics, while its decoded drift fields recovered known lineage and gene programs. By constraining shift magnitude, direction and population shape, the model links predictive accuracy with biological interpretability.

The opportunity

- Distribution-level predictions preserve cellular heterogeneity, including rare non-responsive or resistant states that average endpoints may miss, while trajectories reveal when and how cells enter responsive or resistant states. Together, these capabilities could support virtual-cell models that simulate and prioritize unmeasured interventions and compare predicted responses with patient- and tissue-resolved molecular atlases.

ACKNOWEDGMENT

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